Class 10 Math - Chapter 14: Probability

🌟 Introduction

While Class 9 introduced Empirical (Experimental) Probability based on actual direct observations, Class 10 focuses aggressively on **Theoretical (Classical) Probability**.

Theoretical Probability predicts strictly what *should* happen in an ideal scenario without actually running the physical experiment.

📌 The Fundamental Formula

The theoretical probability of any distinct event E, denoted as $P(E)$, is universally calculated by this core fractional relationship:

$$ P(E) = \frac{\text{Number of outcomes favorable to } E}{\text{Total number of all possible outcomes}} $$

> [!NOTE] The absolute value of any probability must strictly lie flawlessly between 0 and 1 (inclusive). $$ 0 \leq P(E) \leq 1 $$

Important Probability Milestones:

- **Sure Event:** An event that is guaranteed to strictly happen. $P(E) = 1$.
- **Impossible Event:** An event that chemically or mathematically cannot ever happen. $P(E) = 0$.

⚡ Complementary Events

For any given specific event E, the event "**not E**" represents every single other possibility actively occurring. We uniquely denote this strictly as $\bar{E}$ (E bar) or E'.

The sum of the strict probabilities of an event happening and realistically NOT happening is universally exactly 1. $$ P(E) + P(\bar{E}) = 1 $$

Mind Map of Classical Outcomes

```
graph TD;
    Theoretical_Probability-->Coins["Coins (H, T)"];
    Theoretical_Probability-->Dice["Dice (1-6)"];
    Theoretical_Probability-->Cards["Playing Cards (52)"];
    Cards-->Red_26["Red (26)"];
    Cards-->Black_26["Black (26)"];
    Red_26-->Hearts_Diamonds["Hearts, Diamonds"];
    Black_26-->Spades_Clubs["Spades, Clubs"];
```

❌ Common Mistakes

> [!WARNING]
>
> 1. **Card Deck Basics:** Students notoriously forget the precise makeup of a 52-card standard deck. Remember: there are exactly 4 distinct suits (Hearts, Diamonds, Clubs, Spades), natively 13 cards per distinct suit, and universally precisely 12 total Face Cards (Jacks, Queens, Kings).
> 2. **Double Dice Outcomes:** When securely rolling specifically two dice, dynamically multiplying the possibilities yields precisely 36 total structured outcomes, not visually 12. Ensure you dynamically draw the full 6×6 grid to hunt successfully for identical combinations.
> 3. **Leap Year Problem:** A pure non-leap year formally contains precisely 365 days (52 weeks + 1 completely extra standalone day). A massive Leap Year strictly contains accurately 366 days (52 weeks + 2 explicit extra standalone days). The remaining extra standalone days natively decide the unique dynamic probability!

📝 Quick Revision

Common Total Outcomes to Memorize:

- **Tossing 1 coin:** 2 outcomes (H, T)
- **Tossing 2 coins:** 4 outcomes (HH, HT, TH, TT)
- **Tossing 3 coins:** 8 outcomes ($HHH, HHT, HTH, THH, HTT, THT, TTH, TTT$)
- **Rolling 1 die:** 6 outcomes ($1, 2, 3, 4, 5, 6$)
- **Rolling 2 dice:** 36 structured outcomes

Summary

Probability acts flawlessly as the strict mathematical language describing random mathematical uncertainty. By rigorously explicitly writing out the complete detailed Sample Space mathematically first, you physically prevent absolute errors logically occurring strictly later in numerator assignments. Always rigorously check that your final reduced fractional answer dynamically sits perfectly strictly between 0 and 1!